Chapter 6

Diode Applications II: Filters & Regulators

6.1 Introduction

In the previous lecture, we successfully converted AC voltage into DC voltage using rectifiers. However, the output we obtained was **Pulsating DC**.

- **Half-Wave:** Pulses of voltage separated by large gaps of zero voltage.
- **Full-Wave:** Continuous pulses, but the voltage drops to zero 100 times a second (at 50Hz input).

Most agricultural electronic devices (sensors, microcontrollers, GPS modules) require a **smooth, constant DC voltage** (Pure DC), similar to what a battery provides. Feeding pulsating DC into these devices would cause them to malfunction (e.g., a microprocessor would reset 100 times a second!).

To solve this, we need two additional stages in our power supply:

1. **Filter:** To smooth out the pulsations (reduce ripple).
2. **Regulator:** To maintain a fixed voltage level despite changes in load or input.

Learning Objectives

By the end of this chapter, you should be able to:

- Describe the operation of a Capacitor Filter.
- Define Ripple Voltage and calculate the Ripple Factor.
- Explain the Zener breakdown mechanism.
- Design a basic Zener Voltage Regulator circuit.
- Calculate the required Series Resistor (R_S) and Zener Power Rating (P_Z).

6.2 Power Supply Filters

A filter circuit is placed after the rectifier and before the load. Its job is to oppose rapid changes in voltage. The most common type is the **Capacitor Filter**.

6.2.1 The Capacitor Filter Circuit

A large value capacitor (usually Electrolytic, e.g., $470\mu F$ to $4700\mu F$) is connected in **Parallel** with the load resistor R_L.

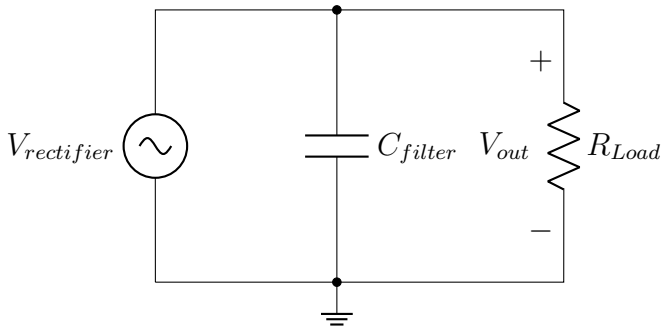


Figure 6.1: Capacitor Filter Circuit connected to Rectifier Output.

6.2.2 Principle of Operation

The capacitor acts like a temporary storage tank for electrons.

Step 1: Charging (Rectifier ON) During the time when the rectifier output voltage is rising towards its peak (V_m), the diode is forward biased. Current flows to

the load AND charges the capacitor. The capacitor charges up to the peak voltage V_m.

Step 2: Discharging (Rectifier OFF) As soon as the rectifier input voltage starts to drop below the peak, the capacitor voltage (V_C) becomes higher than the transformer voltage. The rectifier diodes become reverse biased (turn OFF). Now, the capacitor acts as the source. It discharges its stored energy through the load resistor R_L.

The Result: Because the discharge through the load is relatively slow (large Time Constant $\tau = R_L C$), the voltage drops only slightly before the next charging pulse arrives from the rectifier. This prevents the voltage from ever dropping to zero.

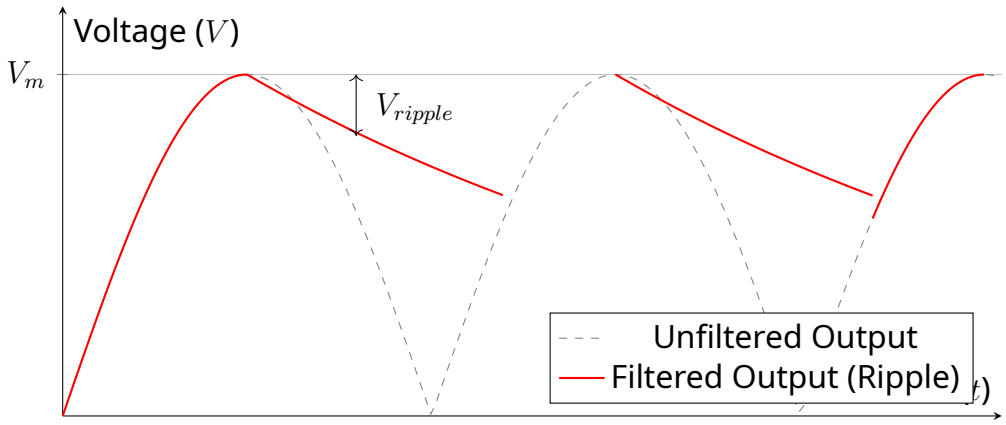


Figure 6.2: Waveform showing the Ripple Voltage caused by charging and discharging.

6.2.3 Ripple Voltage and Ripple Factor

The small fluctuation that remains in the DC output is called **Ripple**.

- **Peak-to-Peak Ripple Voltage ($V_{r(pp)}$):** The difference between the maximum voltage (V_m) and the minimum voltage during discharge.
- **DC Voltage (V_{dc}):** The average value of the filtered output.

$$V_{dc} = V_m - \frac{V_{r(pp)}}{2}$$

Calculating Ripple Voltage: For a Full-Wave Rectifier, the ripple voltage can

be approximated by:

$$V_{r(pp)} = \frac{I_{dc}}{fC} \tag{6.1}$$

Where:

- I_{dc}: The DC load current (Amps).
- f: The ripple frequency (100 Hz for Full-Wave 50Hz input).
- C: The filter capacitance (Farads).

Implication: To reduce ripple (make the DC smoother), we must:

1. Increase the frequency (f).
2. Increase the capacitance (C).
3. Decrease the load current (I_{dc}).

Example 1: Filter Design

A Full-Wave bridge rectifier is fed by a 50 Hz transformer. The load draws a current of 200 mA. Calculate the capacitor value required to keep the peak-to-peak ripple voltage below 1V.

Solution: Given: $I_{dc} = 0.2$ A

$f = 2 \times 50 = 100$ Hz (Full Wave)

$V_{r(pp)} = 1$ V

Using the formula:

$$C = \frac{I_{dc}}{fV_{r(pp)}}$$

$$C = \frac{0.2}{100 \times 1} = 0.002 \text{ F}$$

$$C = 2000\mu\text{F}$$

We would choose the nearest standard value, such as **2200 μF**.

6.3 Zener Voltage Regulators

Even with a good filter, the DC voltage can still change if: 1. The input AC voltage fluctuates (Line regulation issue). 2. The load resistance changes, drawing more

or less current (Load regulation issue).

To fix the voltage to a precise value (e.g., exactly 5.0V), we use a **Voltage Regulator**. The simplest regulator utilizes a **Zener Diode**.

6.3.1 The Zener Diode

The Zener diode is a special silicon diode designed to operate in the **Reverse Breakdown** region.

- **Symbol:** Similar to a standard diode but with "Z" shaped ends on the cathode bar.
- **Forward Bias:** Acts like a normal diode (0.7V drop).
- **Reverse Bias:** It does not conduct until the voltage reaches the **Zener Voltage (V_Z)**. Once reached, it conducts heavily, but the voltage across it remains constantly locked at V_Z.

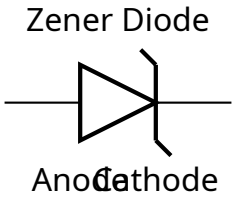


Figure 6.3: Circuit Symbol of a Zener Diode.

6.3.2 Zener Breakdown Mechanism

There are actually two mechanisms, though we loosely call both "Zener breakdown":

1. **Zener Breakdown ($V_Z < 5$V):** In heavily doped diodes, the depletion region is very thin. A strong electric field tears electrons directly from covalent bonds (quantum tunneling).
2. **Avalanche Breakdown ($V_Z > 5$V):** Minority carriers accelerate, collide with atoms, and knock out electrons (impact ionization).

Regardless of the mechanism, the result is the same: **Constant Voltage**.

6.4 Zener Regulator Circuit Design

The Zener diode behaves like a "voltage clamp." To use it, we connect it in **Parallel** with the load, and in **Reverse Bias**. We must also add a **Series Resistor (R_S)**.

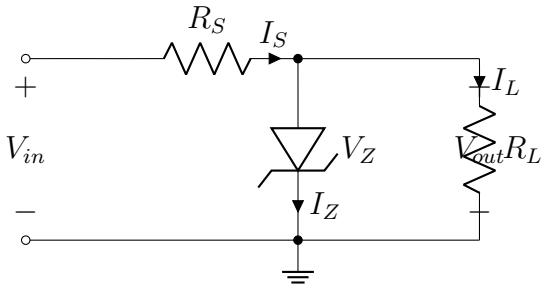


Figure 6.4: Zener Voltage Regulator Circuit.

6.4.1 Operation Principle

- **Kirchhoff's Current Law (KCL):** The total current from the source (I_S) splits into Zener current (I_Z) and Load current (I_L).

$$I_S = I_Z + I_L \tag{6.2}$$

- **Voltage Clamping:** As long as the Zener is in breakdown, the output voltage is fixed:

$$V_{out} = V_Z \tag{6.3}$$

- **Role of R_S:** The series resistor drops the difference between the input voltage and the Zener voltage.

$$V_{R_S} = V_{in} - V_Z \tag{6.4}$$

$$R_S = \frac{V_{in} - V_Z}{I_S} \tag{6.5}$$

6.4.2 Regulation Scenarios

Scenario A: Input Voltage (V_{in}) Increases

1. If V_{in} rises, the total voltage across the circuit increases.

2. The Zener diode "absorbs" the extra current. I_Z increases drastically.
3. Because $I_S = I_Z + I_L$ increases, the voltage drop across R_S increases ($V_{R_S} = I_S R_S$).
4. The Output Voltage remains unchanged ($V_{out} = V_{in} - V_{R_S} = V_Z$).

Scenario B: Load Current (I_L) Changes

1. If we connect a heavier load (decrease R_L), I_L increases.
2. Since V_{in} and R_S are fixed, I_S is constant.
3. Therefore, I_Z must decrease to compensate ($I_Z = I_S - I_L$).
4. The Zener "yields" current to the load while maintaining V_Z.

6.5 Designing the Zener Regulator

To design a reliable regulator, we must ensure the Zener never turns off (current too low) and never burns out (current too high).

6.5.1 Key Parameters

- $I_{Z(min)}$ **/** $I_{Z(knee)}$**:** The minimum current required to keep the Zener in breakdown (typically 1 to 5 mA). If I_Z drops below this, regulation fails.
- $I_{Z(max)}$**:** The maximum current the Zener can handle before overheating.
- $P_{Z(max)}$**:** The power rating of the Zener (e.g., 0.5W, 1W, 5W).

$$I_{Z(max)} = \frac{P_{Z(max)}}{V_Z}$$

Example 2: Zener Regulator Calculation

Design a 10V regulator to power a load that varies between 0 mA and 100 mA. The input voltage is unregulated 15V. Use a Zener diode with $V_Z = 10$V and $I_{Z(min)} = 10$ mA.

Step 1: Calculate Total Source Current (I_S) We need enough current to supply the maximum load (100 mA) AND keep the Zener active (10 mA).

$$I_S = I_{L(max)} + I_{Z(min)}$$

$$I_S = 100 \text{ mA} + 10 \text{ mA} = 110 \text{ mA} = 0.11 \text{ A}$$

Step 2: Calculate Series Resistor (R_S) The resistor must drop the excess voltage ($15V - 10V = 5V$) at the required current.

$$R_S = \frac{V_{in} - V_Z}{I_S} = \frac{15 - 10}{0.11} = 45.45\,\Omega$$

We would choose a standard value, likely **43** Ω (to ensure slightly more current) or **47** Ω.

Step 3: Check Zener Power Rating What happens if the load is disconnected ($I_L = 0$)? All the current flows through the Zener!

$$I_{Z(max)} = I_S = 110 \text{ mA}$$

Power dissipated by Zener:

$$P_Z = V_Z \times I_{Z(max)} = 10 \text{ V} \times 0.11 \text{ A} = 1.1 \text{ Watts}$$

Conclusion: You must buy a Zener diode rated for at least **1.5 Watts** or **2 Watts**. A standard 0.5W Zener would burn out instantly if the load was removed.

6.6 Percentage Regulation

Ideally, the output voltage should never change. In reality, it shifts slightly. We measure the quality of a regulator using two metrics:

6.6.1 Line Regulation

Measures how much V_{out} changes when the *Input Voltage* (V_{in}) changes.

$$\text{Line Reg} = \frac{\Delta V_{out}}{\Delta V_{in}} \times 100\% \tag{6.6}$$

6.6.2 Load Regulation

Measures how much V_{out} changes when the *Load Current* (I_L) changes (e.g., from No Load to Full Load).

$$\text{Load Reg} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% \tag{6.7}$$

Where V_{NL} is voltage at No Load ($I_L = 0$) and V_{FL} is voltage at Full Load. Ideally, this should be 0%.

6.7 Review Questions

Part A: Conceptual Questions

1. Explain the function of a ”Filter” in a DC power supply. What is the most common component used?
2. Define ”Ripple Voltage”. What factors affect the magnitude of ripple?
3. How does a Zener diode differ from a standard rectifier diode in terms of construction and operation?
4. Explain the principle of ”Voltage Clamping” using a Zener diode.
5. What is the purpose of the series resistor (R_S) in a Zener regulator circuit?
6. Define ”Line Regulation” and ”Load Regulation”. Which one measures stability against changing input voltage?
7. Why must a Zener diode be operated in reverse bias? What happens if it is forward biased?
8. Explain what happens to the ripple voltage if the load current is increased.
9. What is the main disadvantage of a simple Zener regulator compared to an IC regulator (like 7805)?
10. Why is it important to consider the ”No Load” condition when selecting the power rating of a Zener diode?

Part B: Advanced Calculations & Analysis

11. **Ripple Calculation:** A full-wave bridge rectifier is fed by a 50 Hz supply. The filter capacitor is 1000μF and the load draws 500 mA. Calculate the peak-to-peak ripple voltage.
12. **Zener Regulator Design:** You need to design a 5V regulated supply from a 12V source. The maximum load current is 50 mA. The Zener diode requires

a minimum of 5 mA to maintain regulation. Calculate the value of the series resistor R_S.

13. **Power Rating:** For the design in Question 12, calculate the maximum power dissipated by the Zener diode if the load is accidentally disconnected. What power rating should the Zener have?

14. **Capacitor Sizing:** You are designing a power supply for a sensitive sensor that requires 12V DC with less than 0.1V ripple. The sensor draws 200 mA. The rectifier is full-wave (100 Hz). Calculate the minimum capacitor value required.

15. **Regulation Analysis:** A Zener regulator has an output of 10.0V at no load. When a full load is connected, the output drops to 9.8V. Calculate the Load Regulation percentage.